GEOPOLITICAL CASE STUDY REPORT

The Geopolitics of Climate Finance, Carbon Markets, and the Green Transition

Climate finance as leverage, carbon pricing as statecraft,
and the CBAM-versus-Global-South confrontation, 2023 to present

October 2026 | Strategic, legal and economic analysis

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Executive Summary

Purpose and Scope

This report examines how climate policy has been transformed from a cooperative, consensus-seeking regime into an arena of structural great-power competition. The central analytical claim is that the instruments through which the green transition is financed, priced and traded (concessional and commercial climate finance, carbon border levies, carbon-credit standards and critical-mineral supply chains) now operate as instruments of statecraft. They allocate market access, confer or withdraw diplomatic legitimacy, and determine which states capture the industrial rents of a post-hydrocarbon economy. The report analyses three interlocking arenas (finance, pricing and markets), a deep-dive case study of the European Union's Carbon Border Adjustment Mechanism (CBAM) and the "climate justice" backlash against it, a structured country-by-country matrix of the Paris Agreement, and a set of policy recommendations oriented toward systemic stability.

The Thesis: The Green Transition Is Being Weaponized

The twentieth-century energy order rested on hydrocarbon chokepoints: the Strait of Hormuz, the Druzhba pipeline, the OPEC quota system and the dollar-denominated petroleum trade. The twenty-first-century order is forming around different chokepoints: refined lithium, cobalt, graphite and rare earths; polysilicon and photovoltaic wafers; battery cathode chemistry; standards for measuring embedded carbon; and the right to certify a tonne of carbon dioxide as "avoided". Whoever controls these nodes acquires leverage over the pace, cost and distributional pattern of everyone else's decarbonization. The thesis of this report is that climate policy is therefore not a separate "environmental" domain that occasionally collides with geopolitics; it is one of the primary domains in which geopolitics is now conducted.

Three observations support the thesis. First, climate finance has become a currency of influence. The Global North's collective pledge of USD 100 billion per year, made at Copenhagen in 2009 and formalized at Cancún in 2010, was meant to be a down payment on trust. It was first reported as met only in 2022, two years late, and with a composition (heavily loan-based, with contested accounting) that the recipient states regard as a breach of the spirit if not the letter of the bargain. The resulting credibility deficit has created a strategic vacuum that China, the BRICS New Development Bank and Gulf sovereign investors are positioned to fill with state-backed financing that carries different conditionalities. Second, carbon pricing has crossed the border. The CBAM, which entered its definitive phase on 1 January 2026, converts the EU's domestic carbon price into a condition of access to a market of roughly 450 million consumers. It is simultaneously an environmental instrument, a defensive industrial measure and a mechanism for exporting regulatory standards. Third, the architecture for trading carbon is fragmenting. Article 6 of the Paris Agreement was designed to create a single, integrity-assured architecture for international carbon cooperation. Its implementation has been slowed by unresolved debates on double-counting, verification and additionality, while national, regional and bilateral systems proliferate with divergent quality thresholds, creating the raw material for rival diplomatic blocs.

Key Friction Points Between the Global North and the Global South

The report identifies six recurring points of friction. (1) **Historical responsibility versus current capability**: the principle of common but differentiated responsibilities and respective capabilities (CBDR-RC) is invoked by developing states against CBAM, which they argue applies a uniform carbon price to unequal starting points. (2) **Finance quantity and quality**: the shift from the USD 100 billion goal to the New Collective Quantified Goal (NCQG) of USD 300 billion per year by 2035, agreed at COP29 in Baku, was welcomed by the North as a tripling and condemned by many in the South as inadequate against estimated needs in the order of USD 1 trillion or more annually. (3) **Debt-creating finance**: a majority of reported public climate finance has historically been provided as loans, compounding fiscal stress in countries that already spend more on debt service than on health or education. (4) **Unilateral trade measures**: Article 3.5 of the UNFCCC states that measures to combat climate change should not constitute arbitrary or unjustifiable discrimination or a disguised restriction on international trade, a provision invoked by India, Brazil, South Africa and China against CBAM. (5) **Technology and mineral value chains**: producers of raw materials (the Democratic Republic of Congo, Indonesia, Chile, Zimbabwe) increasingly seek to capture processing value through export restrictions, which importing states characterise as supply-security threats. (6) **Carbon-market integrity versus market access**: the North emphasizes high-integrity credits, while some developing countries view integrity thresholds as non-tariff barriers that restrict the monetization of their natural capital.

Principal Findings

- The USD 100 billion pledge was a pillar of Northern soft power whose partial delivery has depreciated that power; the NCQG risks repeating the pattern unless private mobilization and grant-equivalent concessionality are made verifiable.
- CBAM is best understood as a hybrid: a legitimate attempt to prevent carbon leakage and protect the integrity of the EU Emissions Trading System (ETS), and an instrument that functions as regulatory extraterritoriality. Its legal durability at the WTO is plausible but not assured, and its political durability depends on the quality of the accommodations (recognition of foreign carbon prices, technical assistance, just-transition partnerships) that accompany it.
- The Global South is not a monolith. India, Brazil, South Africa and China share rhetorical positions against CBAM but differ sharply in exposure, leverage and strategy, and the EU has successfully pursued differentiated engagement through trade agreements, Climate Clubs and bilateral partnerships.
- The United States' formal departure from the Paris Agreement in January 2026, for the second time, has removed the world's largest historical emitter from the architecture and handed the EU and China competing opportunities to define the rules.
- The most plausible trajectory is not rupture but a negotiated, asymmetric fragmentation: multiple overlapping carbon-pricing and climate-finance regimes, with interoperability achieved selectively through mutual recognition, rather than a single universal system.

Structure of the Report

Chapter 1 establishes the macro-geopolitics of the post-hydrocarbon transition. Chapter 2 dissects the weaponization of carbon pricing and the legal and economic mechanics of CBAM. Chapter 3 analyses the fragmentation of carbon markets and the failures of Article 6 implementation. Chapter 4 is the central case study of CBAM and the Global South backlash from 2023 to the present. Chapter 5 provides the structured Paris Agreement matrix. Chapter 6 offers strategic outlook and policy recommendations. A note on sources and data appears at the end of the report: quantitative estimates are cited at the level of precision supported by public sources and should be verified against primary datasets before use in formal submissions.

Chapter 1: The Macro Geopolitics of the Green Transition

1.1 The Post-Hydrocarbon Paradigm Shift

1.1.1 From Energy Commodities to Energy Technologies

The hydrocarbon order was organized around a consumable commodity. Oil and gas are burned; they must be continuously re-supplied, and this continuous dependence is what made them instruments of coercion, from the 1973 oil embargo to Russia's 2022 curtailment of gas flows to Europe. A renewable energy system inverts this logic. Once a solar array, wind farm or battery installation is built, its marginal fuel cost is nil, and the strategic dependence is front-loaded into the supply chain for manufactured capital goods and the minerals embedded in them. The security question therefore shifts from "who will keep supplying fuel?" to "who manufactures, refines and standardises the equipment, and on what terms?" This is a transition from the geopolitics of flows to the geopolitics of stocks and industrial capacity.

The distinction matters because the concentration profile of the new system is, in several segments, more extreme than that of the old. The International Energy Agency (IEA) has documented that the top three producers account for the large majority of global supply across most critical energy minerals, and that refining is more concentrated than mining. China's position is the most consequential: it accounts for roughly 60 to 70 percent of global rare-earth mining and around 90 percent of rare-earth separation and magnet production; around 65 percent of lithium chemical refining; around three-quarters of cobalt refining; more than 90 percent of battery-grade graphite anode production; and upwards of 80 percent of every manufacturing stage in the crystalline-silicon solar supply chain, with polysilicon and wafer stages exceeding 90 percent. These figures vary by year and methodology, but the order of magnitude is not in dispute. No oil-exporting state, including Saudi Arabia in 1973, ever held a comparable share of the downstream processing of the world's energy resource.

1.1.2 Chokepoints in the New Energy Order

Table 1.1 summarises the principal chokepoints and the strategic behaviour they have generated. The behavioural pattern is consistent: concentration invites export controls, export controls invite diversification subsidies, and diversification subsidies invite trade remedies.

Table 1.1: Principal chokepoints in the clean-energy supply chain (approximate shares; sources: IEA and industry data, rounded)

Segment	Approximate concentration	Dominant actor(s)	Observed strategic behaviour
Rare-earth separation and magnets	~90% of separation and magnet output; ~60-70% of mining	China	Export licensing (2025), partly suspended for one year after the US-China leaders' meeting
Lithium refining	~65% of chemical refining	China (mining: Australia, Chile)	Long-term offtake; state participation in Chile and Indonesia
Cobalt	~70% of mining (DRC); ~75% of refining	DRC / China	DRC export suspension and quota system (2025)
Nickel	More than half of mined supply	Indonesia	Ore export ban (2020); forced downstream smelting
Battery-grade graphite (anode)	More than 90%	China	Export licensing (December 2023)
Solar PV chain (polysilicon to module)	80%+ at every stage; 90%+ polysilicon and wafers	China	Overcapacity and low prices; Western tariffs and trade remedies
Battery cells	Roughly 70-80%	China	Technology-export restrictions on selected cathode processes; Western subsidy and tariff response

China's staged use of export licensing is instructive. Controls on gallium and germanium (announced in July 2023, effective August 2023), graphite (December 2023) and antimony (2024) preceded the sweeping rare-earth licensing regime announced in April 2025 and expanded in October 2025. The October measures were partially suspended for one year following the leaders' meeting between the United States and China at the end of that month, which illustrates a defining feature of the new order: chokepoints are not merely used, they are bargained over, and the suspension itself is a bargaining chip. Western responses have been correspondingly legislative and industrial: the US Inflation Reduction Act of 2022, subsequently scaled back by the 2025 budget reconciliation law; the EU's Critical Raw Materials Act of 2024, with benchmarks of 10 percent domestic extraction, 40 percent domestic processing and 25 percent recycling of annual consumption by 2030 and no more than 65 percent dependence on any single third country; and the Minerals Security Partnership and bilateral offtake agreements.

1.1.3 The Fate of Hydrocarbon Exporters

The post-hydrocarbon shift also reorders the fortunes of fossil-fuel states. For exporters with large reserves and low extraction costs, such as the Gulf monarchies, the strategy is to monetize hydrocarbons for as long as demand persists while investing the proceeds in positions in the new system: green hydrogen and ammonia, solar manufacturing, critical-mineral equity stakes and carbon-credit infrastructure. For high-cost or fiscally fragile exporters, such as Nigeria, Angola, Algeria and Iraq, the stranded-asset risk is existential, and these states have an interest in a slower, finance-heavy transition. Their negotiating posture in the UNFCCC process accordingly oscillates between obstruction (resisting fossil-fuel language) and appeals to the principle of just transition funding. The geopolitical reading is that the coalition of states with an interest in a fast transition is smaller than the rhetoric of consensus implies, and that "climate leadership" is frequently correlated with industrial advantage rather than disinterested altruism.

1.1.4 The Industrial Policy Turn

The final structural feature is the return of industrial policy as the organizing logic of the transition. The 2010s assumption that carbon pricing would be the efficient, technology-neutral engine of decarbonization has been supplanted, in the United States, EU and China alike, by direct subsidy, local-content requirements and strategic procurement. The EU's Net-Zero Industry Act and Clean Industrial Deal, the US tax-credit regime, and China's state-guided overcapacity in solar, batteries and electric vehicles are different expressions of the same conviction: that clean technology manufacturing is a strategic sector. For third countries, this creates a squeeze. Developing economies face a world in which the technology they need is produced overwhelmingly by one state, the markets they wish to export to are protected by carbon border measures, and the capital they require is priced at a premium that reflects perceived country risk.

1.2 Climate Finance as the New Foreign Policy Frontier

1.2.1 Origins and Architecture of the USD 100 Billion Pledge

At the 2009 Copenhagen conference, developed countries committed to "a goal of mobilizing jointly USD 100 billion a year by 2020" for developing-country mitigation and adaptation. The Cancún Agreements of 2010 formalised the goal, and the 2015 Paris decision extended it through 2025 and committed to a successor goal above the USD 100 billion floor. Diplomatically, the pledge performed three functions. It was the quid pro quo for developing-country acceptance of a universal, bottom-up regime in which all countries, not only Annex I parties, would take mitigation commitments. It operationalised the principle of differentiated responsibility in material rather than rhetorical terms. And it created a measurable metric against which Northern good faith could be assessed. A pledge with a numeric target and a date is a unique instrument in multilateral diplomacy because it is falsifiable, and its falsification has been politically costly.

1.2.2 Delivery Record and Accounting Disputes

According to figures published by the Organisation for Economic Co-operation and Development (OECD), climate finance provided and mobilised by developed countries reached approximately USD 83.3 billion in 2020 and USD 89.6 billion in 2021, and exceeded the target for the first time in 2022 at approximately USD 115.9 billion. Figure 1.1 shows the trajectory in the OECD series as published at the time of each report; later revisions have altered individual years but not the overall shape.

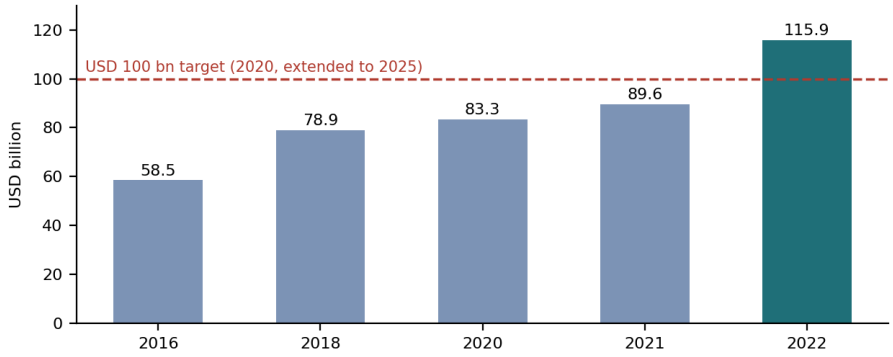


Figure 1.1: Developed-country climate finance provided and mobilised, selected years (USD billion, OECD series as originally published)

The headline figure conceals structural disputes that account for much of the distrust. First, **composition**: public finance, much of it channelled through multilateral development banks, has accounted for the large majority of the total, while "mobilised" private finance, whose attribution methodology is contested, has been a minor component. Second, **instrument**: loans (a large proportion at non-concessional or only partially concessional terms) have constituted roughly two-thirds or more of public climate finance, while

grants have been a minority. Independent analyses, notably those by Oxfam, have argued that when loans are valued by their grant-equivalent and when finance with doubtful climate relevance is excluded, the real support was a fraction of the reported headline, potentially as low as one-third. Third, **thematic balance**: mitigation has dominated, while adaptation, the priority of the most vulnerable states, has received substantially less; the UN Environment Programme's adaptation-gap assessments have repeatedly estimated annual adaptation needs in developing countries at multiples of actual international public adaptation flows. Fourth, **additionality**: much climate finance has been re-labelled development assistance, and there is no agreed definition of "new and additional". Fifth, **geographic distribution**: flows have concentrated in middle-income countries with bankable projects, and the least developed countries and small island developing states have received a disproportionately small share relative to vulnerability.

1.2.3 The NCQG and the Politics of Number-Setting

At COP29 in Baku in November 2024, parties adopted the New Collective Quantified Goal: developed countries are to "take the lead" in mobilizing at least USD 300 billion per year by 2035, within a broader call, addressed to all actors, to scale finance to at least USD 1.3 trillion per year by 2035 through the "Baku to Belém Roadmap". The decision was contested to the point of public rupture. India's delegate described the figure as a "paltry sum" and said that it would not accept the text, while LDC and small-island delegations briefly left negotiating rooms. For analytical purposes three features deserve emphasis. The 2035 horizon defers the larger sums by a decade; the USD 1.3 trillion aspiration is non-binding and diffuses responsibility across public and private, Northern and Southern actors; and the encouragement of voluntary contributions from developing countries (such as China) introduced, for the first time, the principle that major emerging economies should be recognized as potential contributors. This last feature is politically significant because it opens the question of whether China, the world's second-largest economy and largest emitter, should continue to be categorized as a recipient-side actor.

COP30, held in Belém in November 2025 under Brazil's presidency, extended the trajectory without resolving the structural disagreement. The outcome emphasised adaptation finance, including a call to substantially increase it by 2035, and prompted extended debate over trade measures, with Brazil's proposal of regular UN reviews of unilateral trade measures such as CBAM opposed by the EU, which argued that trade disputes belong in trade fora. The political lesson is that the finance and trade agendas, once treated as separate negotiating tracks, are now explicitly linked by Southern negotiators: the quantity of finance is bargained against the stringency of border measures.

1.2.4 Conditionality, Risk Premia and the Cost of Capital

Beyond the headline, the structural financial bias of the system matters more than the pledge. The IEA has repeatedly documented that emerging and developing economies other than China receive a small fraction of global clean-energy investment (on the order of 15 percent) despite accounting for roughly two-thirds of the world's population, and that Africa, with roughly one-fifth of the world's population, attracts only around 2 percent. The cost of capital for utility-scale solar in many developing economies is two to three times that in advanced economies, which can raise the levelized cost of renewable power by tens of percent despite equivalent solar resource. In this context, concessional climate finance is not merely a transfer; it is a risk-pricing intervention whose geopolitical value lies in its ability to unlock private capital that would otherwise price in sovereign risk. Western development finance institutions have deployed blended-finance and guarantee instruments, but the volumes remain insufficient, and conditionalities, including procurement preferences for Western suppliers, governance conditions and, in some cases, coal-retirement milestones, are perceived by recipients as an extension of foreign-policy preferences.

The Just Energy Transition Partnerships (JETPs) illustrate both the promise and the fragility of this model. South Africa's JETP, announced at COP26 in 2021 with an initial USD 8.5 billion, was followed by Indonesia (USD 20 billion, 2022), Vietnam (approximately USD 15.5 billion, 2022) and Senegal (2023). Implementation has been slowed by the predominance of loans over grants, by disagreements about coal-plant retirement schedules, and, in 2025, by the withdrawal of the United States from its JETP commitments, which removed a major prospective contributor and demonstrated how Northern finance pledges are exposed to electoral cycles in donor states. The reliability of the donor is itself an element of the geopolitical equation: a pledge that can be rescinded upon a change of administration confers less leverage than one that cannot.

1.3 The Strategic Vacuum: China's Green Belt and Road and the BRICS Alternative

1.3.1 The Vacuum Defined

A strategic vacuum arises when the supply of a good that states value falls short of demand, and a potential supplier with different preferences exists. The shortfall in Northern climate finance, combined with the withdrawal or reduction of US contributions, the diversion of European budgets toward defence and energy security after 2022, and the cutbacks in several donor states' official development assistance in 2024 and 2025, has expanded the gap between what developing countries need and what traditional donors supply. The vacuum is not only financial; it is also normative. Developing states have a preference for finance that does not carry Western governance conditionality, that is delivered quickly, and that is attached to tangible infrastructure rather than policy reform. These are precisely the attributes that Chinese policy banks have historically offered.

1.3.2 China's Green Belt and Road Initiative

China's Belt and Road Initiative (BRI), launched in 2013, has financed several hundred billion dollars of infrastructure, with the energy sector as a large component. For much of the 2010s, that portfolio was heavily weighted toward coal and hydropower; the 2021 announcement by President Xi Jinping that China would not build new coal-fired power projects abroad marked a formal pivot. Since then, the portfolio has shifted markedly toward solar, wind, grid and storage assets, together with the manufacturing capacity that supplies them. The institutional vehicle is the "Green Development" strand of the BRI, supported by the BRI International Green Development Coalition, the Green Investment Principles and, increasingly, state-backed guidelines on overseas investment. The political economy is distinctive in three respects.

First, **vertical integration**: Chinese firms are simultaneously financiers' counterparties, equipment suppliers and engineering contractors, which can compress delivery times and reduce costs, but which also generates lock-in to Chinese technology standards, grid-code conventions and maintenance supply chains. Second, **manufacturing outbound investment**: since about 2023, Chinese solar, battery and electric-vehicle producers have established factories in Southeast Asia, the Middle East, North Africa, Latin America and sub-Saharan Africa. This allows them to serve local markets and to circumvent Western tariffs and local-content rules aimed at Chinese-origin goods. Third, **South-South framing**: China presents its engagement under the banner of South-South cooperation, in which it positions itself as a developing country that offers assistance rather than a donor with obligations, and has made voluntary pledges (including multi-billion-yuan South-South climate cooperation funds and the supply of low-carbon equipment) that sit outside the formal "developed-country" accounting framework. This positioning allows China to claim leadership of the developing world while avoiding the legal obligations that attach to Annex II parties.

The commercial dimension is substantial. The overwhelming dominance of Chinese-manufactured panels, inverters and increasingly batteries means that the practical choice facing a developing-country utility in 2026 is frequently between Chinese equipment financed by Chinese lenders and a considerably more expensive alternative. In 2024 and 2025, a surge of low-cost Chinese solar exports to Pakistan, Africa and the Middle East demonstrated how price, rather than policy, can accelerate energy transitions in states that were never primary recipients of Northern climate finance. The result is a form of structural influence that does not require formal alliance: partner states acquire an interest in the continued competitiveness of Chinese suppliers, in the stability of Chinese financing, and in the diplomatic positions that safeguard both.

1.3.3 Risks and Limits of the Chinese Model

It would be analytically careless to treat the Chinese model as an unambiguous strategic success. The BRI has been associated with debt-sustainability concerns in several partner states, and Chinese lenders have undertaken renegotiations and, in some cases, write-downs. Domestic overcapacity in clean-technology manufacturing has generated price deflation, trade-remedy actions in the EU, United States, India, Brazil and others, and political backlash in markets such as South Africa and Turkey where local industry has been displaced. China's own carbon-peaking trajectory remains contested: its 2035 national contribution, announced in September 2025, targets a reduction of 7 to 10 percent from peak levels, which many observers regard as modest relative to China's technological capacity. Finally, China's refusal to accept the classification of a "developed country" for the purposes of climate finance exposes it to a legitimacy challenge if its relative economic weight continues to grow.

1.3.4 The BRICS New Development Bank and Institutional Alternatives

The New Development Bank (NDB), established by the BRICS states in 2014 and operating from Shanghai since 2015, was conceived as an alternative to Bretton Woods institutions with a governance structure of equal voting shares among founders and a lending philosophy that avoids policy conditionality. Under the presidency of former Brazilian President Dilma Rousseff, the NDB has expanded its membership to include Bangladesh, the United Arab Emirates, Egypt and Algeria, with additional states in the pipeline, and has committed a large share of its portfolio to sustainable infrastructure, including renewable energy and water. It has also pursued local-currency lending to reduce reliance on the US dollar, a strategic objective directly connected to de-dollarization narratives. The Asian Infrastructure Investment Bank (AIIB), established in 2016, shares part of this mandate, although its membership includes numerous Western states.

The NDB's constraints should be recognised. Its balance sheet is small relative to the World Bank Group and the regional development banks, its AAA/AA rating access has been complicated by sanctions-related complications with the Russian portfolio (it suspended new Russian transactions in 2022), and its potential for scale is bounded by the capital contributions of its shareholders. Its strategic value is therefore less in volume than in institutional precedent: it demonstrates that a Southern-led multilateral lender is feasible, that members can borrow in their own currencies, and that alternative governance norms can attract members. The BRICS grouping's 2024 enlargement and the emergence of BRICS-level cooperation on carbon markets and climate finance (including the presentation, under the 2024 Russian chair, of a BRICS carbon-markets partnership, and Brazil's 2025 chairship, which prioritised climate finance and a "climate-justice" framing) indicate that the bloc intends to use climate policy as a vehicle for cohesion even where its members have divergent interests in practice. India's reluctance to embrace overt anti-Western alignment, Brazil's hosting of COP30 as a bridge-builder, and China's strategic competition with the United States produce a bloc that is cohesive on process and divergent on substance.

1.3.5 Strategic Assessment: Competition for the Middle

Three conclusions follow. First, the contest is not primarily for the allegiance of the world's poorest countries, which lack the fiscal space to absorb large loans and have limited market leverage; it is for the "swing states" (Indonesia, Brazil, India, South Africa, Vietnam, Turkey, Saudi Arabia, the UAE and Mexico) whose critical minerals, consumer markets and diplomatic weight give their alignment disproportionate value. Second, the competition is multidimensional: financing terms, technology access, market access and standard-setting authority are all instruments, and states with a diversified menu (as the EU's Global Gateway programme, with a headline mobilization of EUR 300 billion for 2021 to 2027 and subsequent upward revisions, attempts to provide) are better positioned than those with a single instrument. Third, the vacuum is not static. It is filled partly by Chinese capital, partly by Gulf sovereign wealth, partly by multilateral development banks undergoing reform under the Bridgetown agenda and the Group of 20 capital-adequacy reviews, and partly by the countries themselves, as India and Brazil develop domestic green-finance taxonomies and carbon markets. The relevant question for Western policymakers is therefore not whether the vacuum will be filled, but on what terms, and whether the terms will be compatible with the interests and norms they wish to preserve.

Chapter 2: The Weaponization of Carbon Pricing and Green Protectionism

2.1 From Domestic Price Signal to Border Instrument

2.1.1 The Leakage Problem and Its Political Economy

A carbon price imposed by one jurisdiction raises production costs for its energy-intensive, trade-exposed industries. If competitors elsewhere do not face an equivalent price, production and emissions may migrate ("carbon leakage"), undermining both environmental effectiveness and domestic political support. The EU Emissions Trading System, launched in 2005, historically addressed the problem by allocating free allowances to exposed sectors, a form of implicit subsidy that blunted the incentive to decarbonize. As the EU raised its 2030 target to a 55 percent net reduction and the ETS price rose to the range of EUR 60 to 100 per tonne in 2022 to 2025, the free-allocation model became fiscally and politically untenable: it was criticized by environmentalists for diluting the price signal and by industry for creating uncertainty. CBAM resolves this dilemma by replacing free allocation with a border charge that equalises the carbon cost between domestic production and imports.

The political economy that produced CBAM was therefore domestic at least as much as international. The legislation was the product of three coalitions: environmental actors demanding higher ambition, industrial actors demanding protection from foreign competitors with lower carbon costs, and fiscal actors who welcomed a new revenue stream. A structure that satisfies all three has an attractive internal equilibrium, but its external consequences were secondary in the design phase, which partly explains the intensity of the external reaction.

2.1.2 The Concept of Regulatory Extraterritoriality

The term "weaponization" requires analytical discipline. Not every trade measure with environmental purpose is coercive. We use it here in the following sense: a measure is weaponized where (a) it conditions access to a large market on the adoption of the regulating state's domestic standards; (b) the costs of non-adoption fall substantially on third countries that did not participate in the standard's design; and (c) the regulating state anticipates, and values, the secondary effect of inducing foreign policy convergence. By these criteria CBAM clearly satisfies (a) and (b). Criterion (c) is explicit in EU policy documents, which describe CBAM as intended to "encourage cleaner industrial production in non-EU countries", and in the prevailing academic literature on the "Brussels Effect", in which large, regulation-intensive markets unilaterally export their standards because exporters find it more efficient to comply globally than to maintain separate production lines.

The strategic logic is that of a climate "club" in the sense developed by William Nordhaus in 2015: a coalition imposes tariffs on non-members to induce participation. The EU's version differs in that it is not a formal club with membership rules but a unilateral measure that functions as a club in practice, offering exemption (as with Iceland, Liechtenstein, Norway and Switzerland, which are integrated with the EU ETS) or deduction (for carbon prices paid elsewhere) to those who adopt equivalent systems.

2.2 The Mechanics of CBAM

2.2.1 Legal Basis and Scope

CBAM was established by Regulation (EU) 2023/956 as part of the "Fit for 55" package. Its initial scope covers cement, iron and steel, aluminium, fertilisers, electricity and hydrogen, selected because of their high emissions intensity and exposure to leakage. These sectors represent a large proportion of emissions from ETS-covered industries; the Commission has stated that, once free allocation is fully phased out, CBAM will capture a majority of the emissions in the ETS sectors it covers. The scope includes both

direct emissions and, for some products, indirect emissions from electricity consumption, and extends to selected downstream products such as screws, bolts and certain steel and aluminium articles. In December 2025 the Commission proposed a further extension to a wider set of downstream goods and an anti-circumvention provision, reflecting a recognition that the original scope created an incentive to shift exports toward further-processed products outside the regulation.

2.2.2 The Two Phases

The transitional phase ran from 1 October 2023 to 31 December 2025. During this period importers were required to submit quarterly reports on the embedded emissions of their imports, but no financial adjustment applied. The aim was to build a methodological and administrative base. The definitive phase began on 1 January 2026. Following the “Omnibus” simplification package adopted in 2025, three significant modifications apply. First, a single de minimis threshold of 50 tonnes of imported CBAM goods per importer per year replaced the earlier EUR 150-per-consignment exemption; the Commission estimates that this removes a large majority of importers (around 90 percent) from the scope while retaining about 99 percent of embedded emissions. Second, the sale of CBAM certificates has been deferred to 1 February 2027; certificates to cover the 2026 import year are acquired in 2027 at a price based on the 2026 quarterly average of EU ETS auction prices, and the first annual declaration for 2026 imports is due by 30 September 2027. Third, the requirement to hold certificates covering a share of embedded emissions at the end of each quarter has been reduced from 80 percent to 50 percent. The first official CBAM certificate price, for the first quarter of 2026, was published by the Commission as EUR 75.36 per tonne of CO2e.

Only authorised CBAM declarants may import covered goods above the threshold; unauthorised importers face import restrictions and penalties. The financial implications are therefore effectively deferred but legally fixed: obligations accrue on 2026 imports, and exporters and importers have been calibrating contracts accordingly since the start of the year.

2.2.3 The Calculation: Embedded Emissions, Benchmarks and the CBAM Factor

The financial obligation for each importer is, in simplified form, a function of four variables: the embedded emissions of the imported goods; the EU ETS price; the share of the obligation offset by the phase-out of free allocation (the “CBAM factor”); and any deduction for a carbon price effectively paid in the country of origin. In schematic form, the obligation equals the embedded emissions, minus the free allocation that EU producers would have received for the equivalent product (adjusted by the CBAM factor), multiplied by the ETS price, minus the carbon price paid abroad.

The CBAM factor is the mechanism through which the border charge is phased in as free allocation is phased out. The schedule is: 2.5 percent in 2026, 5 percent in 2027, 10 percent in 2028, 22.5 percent in 2029, 48.5 percent in 2030, 61 percent in 2031, 73.5 percent in 2032, 86 percent in 2033, and 100 percent from 2034. Figure 2.1 plots this schedule. The central point for strategic analysis is the back-loaded profile: in the first years, the financial burden is modest, giving exporters time to adapt, but from 2030 the schedule steepens rapidly, so that by the early 2030s the full ETS price applies to imported carbon content. Exporters who postpone decarbonization investments until the burden becomes acute face a compressed adjustment window.

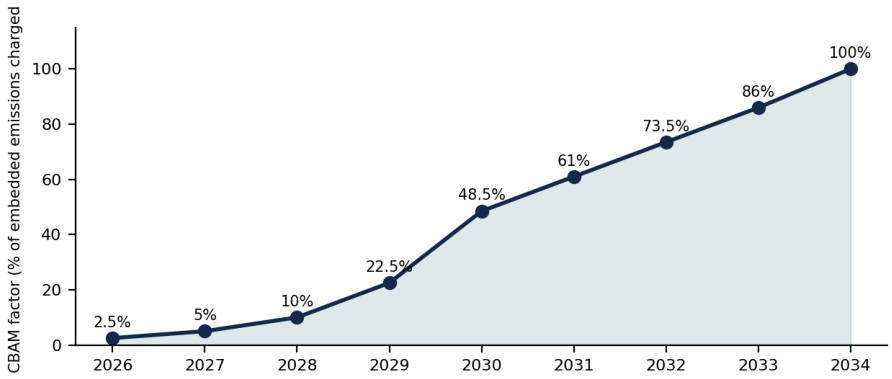


Figure 2.1: CBAM factor schedule, 2026-2034 (share of obligation phased in as EU free allocation is withdrawn)

2.2.4 Default Values and the Verification Problem

Where an exporting installation cannot demonstrate its actual emissions through an accredited verifier, the importer must use default values published by the Commission. Those defaults are deliberately conservative: they are based on the average emissions intensity of the worst-performing producers in the exporting country, with a mark-up that the regulation phases in over time. The incentive structure is clear. Producers who invest in monitoring, reporting and verification (MRV) infrastructure and who can demonstrate lower-than-default emissions benefit; those who do not are penalized by the default. The strategic effect is that the EU has, in effect, set a global MRV standard for covered sectors, because it controls the methodology by which emissions are measured. A producer in Odisha or Gauteng who wishes to export steel to Rotterdam must adopt EU-compatible measurement boundaries,

installation-level data collection and third-party verification regardless of whether the domestic regulator requires it. This is the plainest operational example of regulatory extraterritoriality.

An illustration conveys the magnitudes. For steel produced by a blast-furnace route with embedded emissions in the range of 2.0 to 2.4 tonnes of CO2e per tonne of product, a price of EUR 75 per tonne translates into an unadjusted charge of EUR 150 to 180 per tonne of steel; at the 2026 CBAM factor, after deductions for the free allocation the EU producer would have received, the effective charge is a small fraction of that. By 2034, at 100 percent phase-in, the same product carries a charge of the order of EUR 150 or more per tonne at constant prices, which for commodity steel with thin margins is a material share of the export price. These figures are illustrative; actual charges depend on product-specific benchmarks, the evolution of ETS prices and any foreign-price deductions.

2.2.5 Recognition of Foreign Carbon Pricing

Article 9 of the regulation allows a deduction for the carbon price "effectively paid" in the country of origin. This is the principal channel through which the EU can reward countries that adopt their own carbon pricing, and it is the mechanism that gives non-EU governments a direct interest in designing domestic systems that the EU will recognise. As of mid-2026, the Commission's draft implementing measures reportedly name jurisdictions with established systems, including the United Kingdom, China and California, and do not yet name India, whose Carbon Credit Trading Scheme (CCTS) is still being phased in. During the January 2026 conclusion of the EU-India free trade negotiations, India's commerce secretary stated that there was an understanding that India's future carbon trading system would be recognised under CBAM, whereas Commission representatives were explicit that the agreement provides no exemption from CBAM and no more favourable treatment than other partners. These two statements are reconcilable only on the reading that recognition is a technical administrative process rather than a political concession, which is exactly the position the EU wishes to maintain for legal reasons.

2.3 Economic Coercion or Environmental Protection? Competing Interpretations

2.3.1 The Case That CBAM Is Legitimate

Defenders advance four arguments. (1) **Non-discrimination**: CBAM applies the same carbon cost to EU producers and importers, and thus is a neutral border-tax adjustment of the kind long accepted for value-added taxes and excise duties. (2) **Environmental effectiveness**: it prevents leakage and thereby protects the integrity of an ambitious domestic policy; without it, the EU would either need to retain free allocation (which weakens decarbonization incentives) or accept industrial relocation. (3) **Incentive alignment**: it rewards the adoption of carbon pricing elsewhere, and so encourages the diffusion of climate policy: a significant number of countries, including Turkey, Brazil, India, Indonesia and Vietnam, have accelerated design of domestic carbon markets in explicit reference to CBAM. (4) **Revenue retention**: where a carbon price is paid, it is preferable for the revenue to be retained in the exporting country (through domestic pricing) than to flow to the EU treasury, an argument that gives exporting governments a financial reason to introduce their own price.

2.3.2 The Case That CBAM Is Coercive

Critics respond on five grounds. (1) **Distributive asymmetry**: the charge falls most heavily on states with carbon-intensive industry, higher grid emissions intensity and limited fiscal capacity to pay for abatement, which are typically developing countries. Estimates by UNCTAD and several think tanks suggest that the emissions-reduction effect of CBAM is small (on the order of fractions of a percent of global emissions) while the trade impact on specific exporters is concentrated; Mozambique's aluminium sector, Zimbabwe's ferro-alloys and India's steel and aluminium exports to the EU are repeatedly cited as exposed. India's exports of covered goods to the EU have been estimated by Indian think tanks at roughly one-fifth to one-quarter of its exports in those categories. (2) **Historical responsibility**: CBAM implicitly treats all producers as equally accountable for current emissions intensity, overlooking the fact that today's high-income states built their industrial base without carbon costs. (3) **Capacity asymmetry**: the MRV requirements impose fixed compliance costs that are disproportionately burdensome for small producers. (4) **Design opacity**: the EU controls default values, benchmarks, recognition decisions and amendments, with limited consultation with affected exporters. (5) **Use of market power**: the EU's market size makes exclusion an effective threat, and the secondary objective of inducing foreign policy convergence is publicly acknowledged.

2.3.3 Assessment

An evidence-based assessment concludes that both accounts are partly right and that the apparent contradiction dissolves once the measure's *intent* and *effect* are distinguished. The legal intent is environmental and non-discriminatory. The practical effect is to transfer regulatory authority over the measurement and pricing of industrial emissions from producing states to the EU for the portion of production destined for the EU market. This is not unique to CBAM: US sanctions regimes, GDPR and EU chemicals legislation all work in the same way. What is distinctive is that climate policy carries a particular moral claim, the justification of CBAM as necessary to avert planetary harm, which makes critics' rejection of the measure appear, to Northern audiences, as obstruction. The central analytical insight is that legitimacy, not legality, is the binding constraint on CBAM's durability.

2.4 Beyond Europe: The Diffusion of Border Carbon Adjustments

2.4.1 The United Kingdom, the United States and Others

The United Kingdom has legislated its own CBAM, scheduled to apply from 2027, with scope similar to the EU's. The UK-EU reset of 2025 included a commitment to explore linking the two emissions trading systems, which if completed would allow mutual exemption. India's government has publicly reserved the right to respond to the UK measure and, given the UK-India comprehensive trade agreement signed in 2025, has sought to manage the interaction through negotiation. In the United States, Congressional proposals such as the Foreign Pollution Fee Act and the Clean Competition Act have circulated, but the policy direction under the current administration is not carbon pricing; instead it relies on tariffs applied on grounds of trade and national security. The 2025 US-EU framework statement referenced possible flexibilities in CBAM for US small and medium-sized enterprises, suggesting that Washington regards CBAM chiefly as a trade irritant to be bargained over. Canada, Japan, Australia, Taiwan and others have studied or begun to design border adjustment mechanisms. The emerging picture is of multiple regimes whose interoperability is uncertain.

2.4.2 A Taxonomy of Green Protectionism

To distinguish CBAM from other green trade measures, it is useful to classify the instruments in current use. The categories overlap, but the taxonomy clarifies the strategic landscape.

Table 2.1: A taxonomy of green protectionism

Instrument	Examples	Mechanism	Primary geopolitical effect
Border carbon adjustment	EU CBAM; UK CBAM (planned 2027)	Carbon cost levied on embedded emissions of imports	Exports the regulator's measurement and pricing standards
Clean-technology subsidies and local content	US tax credits (as amended in 2025); EU Net-Zero Industry Act	Credits, procurement preferences, domestic-content tests	Industrial reshoring; discrimination claims by exporters
Trade remedies on clean technology	Anti-dumping and countervailing duties on solar, EVs (EU, US, India, others)	Duties on Chinese-origin goods	Diversification; retaliation and supply-chain re-routing
Export controls on inputs	China (rare earths, graphite); Indonesia (nickel); DRC (cobalt)	Licensing, quotas, bans	Leverage over downstream manufacturers
Product-footprint and due-diligence rules	EU Deforestation Regulation; EU Batteries Regulation	Disclosure, footprint thresholds, due diligence	Standard-setting authority over global supply chains
Credit-eligibility rules	EU 2040 international-credit rules; CORSIA	Quality criteria and host-country authorisation	Gatekeeping of Southern credit supply

2.4.3 The WTO Legal Landscape

The legal assessment of CBAM under World Trade Organization law turns on the interplay of several provisions. Article III:2 of the General Agreement on Tariffs and Trade (GATT) requires national treatment for internal taxes on imported products, and the EU argues that CBAM is an internal charge applied to imports in a manner equivalent to the domestic ETS obligation. Article I (most-favoured-nation treatment) is engaged by the differential treatment of imports based on the carbon price of the country of origin and by the exemption of countries linked to the ETS: the EU must argue that products from countries with and without carbon pricing are not "like" for this purpose, or that the differentiation is justified under Article XX. Article XX(b) (measures necessary to protect human, animal or plant life or health) and XX(g) (measures relating to the conservation of exhaustible natural resources) provide possible justification, subject to the chapeau's prohibition on arbitrary or unjustifiable discrimination and disguised restriction. The Appellate Body's jurisprudence in *US-Shrimp* held that unilateral measures can be justified under XX if they incorporate flexibility to take into account the conditions prevailing in exporting countries, which suggests that the design features of CBAM (deductions for foreign carbon prices, case-by-case treatment of installation data) are legally significant. Russia's request for WTO consultations on CBAM in 2025 initiated the first formal dispute. Given the paralysis of the Appellate Body since 2019, a panel report could be appealed "into the void", which means that the legal contest may serve as a political signalling forum rather than a definitive adjudication. This structural feature, discussed in Chapter 4, shapes the strategies of all parties.

Chapter 3: The Fragmentation of Global Carbon Markets and Article 6 Failures

3.1 The Architecture That Was Promised

3.1.1 Article 6 of the Paris Agreement

Article 6 of the Paris Agreement establishes three approaches to international carbon cooperation. Article 6.2 enables bilateral and plurilateral arrangements in which countries transfer "internationally transferred mitigation outcomes" (ITMOs), subject to "corresponding adjustments" to avoid the same reduction being counted by both the selling and the buying country. Article 6.4 creates a centralised crediting mechanism, the Paris Agreement Crediting Mechanism (PACM), supervised by a UN body and designated as the successor to the Kyoto Protocol's Clean Development Mechanism (CDM). Article 6.8 provides for non-market approaches, a framework for cooperation not involving the transfer of credits. In design, Article 6 was intended to reduce global abatement costs, with the often-cited estimate that cooperation could lower the cost of implementing national contributions by as much as half, and to channel private finance to developing countries.

The negotiation of the rulebook took six years. The detailed rules for Article 6.2 and 6.4 were agreed in principle at COP26 in Glasgow in 2021 and completed at COP29 in Baku in 2024, where parties adopted standards for methodologies and removals under the PACM. The delay is significant: a mechanism that had been intended to operate from 2021 became operational in earnest only from 2025, by which point national contributions were already being implemented through parallel arrangements.

3.1.2 The Strategic Promise

Article 6 held three strategic promises. It would provide a **credible quality floor**, ensuring that credits traded internationally satisfied common standards. It would offer a **monetization pathway** for developing countries' mitigation potential, particularly in forestry, cookstoves and renewable energy. And it would offer a **neutral venue** under UNFCCC auspices in which no single state or bloc controlled the rules. Each promise has been compromised.

3.2 The Obstacles

3.2.1 Double Counting and Corresponding Adjustments

The principle of corresponding adjustment seems simple: if Country A sells an ITMO to Country B, A adds the transferred tonnes to its own emissions inventory and B subtracts them. In practice, three difficulties arise. First, the **sovereign cost** of the adjustment: a host country that authorises the export of reductions foregoes the ability to count them toward its own national contribution, which is rational only if the price received exceeds the opportunity cost of future compliance, and which requires states to forecast their own marginal abatement costs decades in advance. Second, **registry coordination**: ITMOs may be issued, transferred, used and cancelled across registries (national, UN, voluntary) whose technical interoperability is incomplete. Third, the **voluntary-market interface**: credits issued under voluntary-market standards and used for corporate claims have traditionally not received corresponding adjustments. Governments increasingly require them for credits to be recognised as "Paris-aligned", but the voluntary market's position is contested: an unadjusted credit may be sold to a company for use in a claim of "carbon neutrality" while the host country counts the same reduction toward its own contribution, which is double counting in substance if not in legal form.

The most developed responses to this problem are in the compliance-linked segment. Singapore, which allows covered entities to use international credits for up to 5 percent of their taxable emissions, has concluded implementation agreements with a series of partners requiring authorization and corresponding adjustments. Switzerland, Japan (through the Joint Crediting Mechanism) and Sweden have pursued bilateral deals, and the first formal ITMO transfers took place in 2024 and 2025. Volumes remain small relative to the scale of the voluntary market, and some host countries (Ghana, Kenya and others) have adopted regulations imposing revenue shares, in-country benefit-sharing and authorisation requirements that cut the net price to project developers.

3.2.2 Verification, Baselines and Additionality

The integrity of any crediting system depends on the credibility of its counterfactual: what would emissions have been without the project? The history of the CDM and voluntary market shows persistent problems with inflated baselines. A body of academic work, including large-scale analyses of voluntary-market REDD+ (reducing emissions from deforestation and forest degradation) projects, has found that a majority of credits issued did not correspond to real reductions, and investigations published in 2023 in the Guardian, Die Zeit and SourceMaterial concluded that a high proportion of rainforest credits certified by the largest standard-setter were "phantom credits". Similar concerns attach to certain renewable-energy credits (where projects would have been built regardless) and to cookstove credits (where usage and fuel-displacement assumptions are overstated). These findings contributed to

a collapse in the voluntary market: transactions that had grown beyond USD 2 billion in 2021 and 2022 contracted sharply in 2023 and 2024 as buyers withdrew.

Remedial institutions have emerged: the Integrity Council for the Voluntary Carbon Market (ICVCM) and its Core Carbon Principles; the Voluntary Carbon Markets Integrity Initiative (VCMI); the Science Based Targets initiative's position on offsets; and the International Civil Aviation Organization's CORSIA scheme, which established eligibility criteria for aviation offsets and includes a requirement for host-country letters of authorization. The PACM's Supervisory Body has adopted methodologies and standards, but its capacity to review and approve methodologies at scale is untested, and its transition rules for legacy CDM projects are controversial.

3.2.3 Greenwashing, Reputational Risk and Legal Liability

Buyers' reluctance is amplified by regulation. The EU's directive on empowering consumers for the green transition, adopted in 2024, restricts claims of climate neutrality based on offsetting; the UK, Australia and several US states have taken enforcement actions; and the EU's Green Claims proposals, though subject to political turbulence in 2025, signalled the direction of regulatory travel. The consequence is that demand from Northern corporates has moved toward fewer, higher-quality credits and toward removals rather than avoidance. This benefits technology-intensive removals (direct air capture, biochar, enhanced weathering) concentrated in developed countries, while reducing demand for the nature-based and cookstove credits that constitute the principal supply from developing countries. The integrity agenda therefore has a distributive side effect, which Southern governments increasingly describe as a form of market closure, and which the next section shows is feeding a search for alternative venues.

3.2.4 The Unresolved Question of the EU's 2040 Target

The relationship between Article 6 and the EU's own climate policy highlights the stakes. In 2025 the EU agreed a 2040 target of a 90 percent net reduction in emissions relative to 1990, with the possibility of using up to 5 percent in international credits from 2036 subject to Article 6 standards. Although the mechanism's design is still being defined, it signals that Europe, which long resisted offsetting in its domestic target, accepts that cost-effectiveness may require access to external abatement. This is a strategically important validation of Article 6 but also creates a new source of influence: the EU's definition of eligibility will determine which host countries and project types can access the demand, in effect making Brussels a quality regulator for international carbon assets.

3.3 The Rise of Parallel and Lower-Standard Systems

3.3.1 Why Fragmentation Is Occurring

Fragmentation is not simply the failure of Article 6; it is the predictable consequence of unequal bargaining positions. States with strong domestic demand for credits (the EU, the UK, Japan, South Korea, Singapore) want high integrity standards to protect their own regulatory credibility. States with large supply potential want lighter standards and higher prices. States that distrust the UNFCCC process, or that wish to avoid the sovereign accounting consequences of corresponding adjustments, prefer bilateral or club arrangements. The easiest path for each is to design its own system. The result is a patchwork in which at least four layers coexist: national compliance markets (EU ETS, UK ETS, China's national ETS, Korea's K-ETS, and the emerging systems of India, Brazil, Indonesia and Turkey); the Article 6.2 bilateral channel; the Article 6.4 PACM; and voluntary-market and regionally branded systems.

3.3.2 China's National ETS and the CCER

China's national emissions trading system, launched in July 2021 for the power sector, is the largest in the world by covered emissions (over 5 billion tonnes of CO2, roughly 40 percent of national emissions). It uses an intensity-based allocation with benchmark-driven free allowances, so that it does not impose an absolute cap and the carbon price (typically in the range of CNY 60 to 100 per tonne) is far below that of the EU. In 2025 the system was announced to extend to steel, cement and aluminium, which is directly relevant to CBAM: a Chinese carbon price paid by exporters would be deductible under Article 9 provided it is "effectively paid" in the CBAM sense. The reintroduced China Certified Emission Reduction (CCER) scheme, relaunched in 2024 after a seven-year suspension, offers a domestic offset channel with Chinese methodologies. The strategic significance is twofold. China is creating a carbon-market infrastructure that is nominally lower in price and different in method from that of the EU, and it can use the scale of its market to define regional standards. And by building a domestic price with an explicit eye toward CBAM recognition, China is converting a potential trade penalty into an administrative negotiation.

3.3.3 India, Brazil and the Emerging Compliance Markets

India's Carbon Credit Trading Scheme, based on the 2022 amendment to the Energy Conservation Act, relies on greenhouse-gas intensity targets for covered entities, with those over-performing eligible to receive tradeable credits; sectoral targets have been notified for industries including aluminium, cement, chlor-alkali, and pulp and paper, with compliance phasing in from 2026. Brazil enacted a law establishing the Brazilian Greenhouse Gas Emissions Trading System (SBCE) in December 2024, which will have a regulatory and implementation phase of several years. Indonesia launched a carbon exchange in 2023 and issued regulations on

international trade of carbon credits in 2025; Turkey is preparing a national ETS in anticipation of CBAM. In every case the stated motivation includes "retaining carbon revenue domestically rather than paying it to the EU". These systems are not interoperable by design: they differ in coverage, use intensity rather than absolute caps, and in several cases lack independent verification, so their recognition by the EU, and their mutual compatibility, is an open question.

3.3.4 Gulf Platforms and the African Carbon Initiative

A different pattern appears in the Gulf and Africa. Saudi Arabia's Regional Voluntary Carbon Market Company (RVCMC), launched in 2022 with Public Investment Fund backing, and the Abu Dhabi Global Market's carbon-credit infrastructure position the Gulf as a hub for carbon-credit trading, in line with these states' interest in monetizing both hydrocarbon assets and abatement. The African Carbon Markets Initiative (ACMI), launched at COP27, aims to expand African credit production dramatically by 2030; critics note that such ambitions face the same integrity questions as other voluntary-market programmes and that the economics of benefit-sharing remain contested. African governments are also asserting sovereignty over credit authorisation: Kenya, Zimbabwe, Nigeria and Tanzania have adopted or proposed rules on revenue shares, and Zimbabwe's 2023 decision to claim a large revenue share and require government authorisation from the Kariba REDD+ project illustrates the willingness of host states to reallocate rents. This "carbon nationalism" mirrors resource nationalism in mining and hydrocarbons, and the sectors share a common political logic: host states seek to capture a greater portion of the value of assets located within their territory.

3.3.5 Carbon Markets as Diplomatic Blocs

The strategic risk arises when market design is used to establish blocs. The bilateral Article 6.2 agreements that Singapore, Japan, Switzerland and Sweden have signed create "carbon corridors" linking buyers with preferred hosts, with the buyer's standards embedded in the partnership. The Japanese JCM, which has partnered with more than twenty countries, is explicitly connected to Japan's technology-export policy. The BRICS carbon-market partnership introduced in 2024 is an attempt to create a venue in which Southern producers and large emerging-market compliance systems can coordinate methodologies and potentially accept each other's credits, with the political subtext of reducing dependence on Western-controlled standards. China is positioned as a potential hub: its market scale, state-owned financial institutions, and the digital infrastructure of its national registry offer the possibility of a regional standard that competes with the Western standard and which could be offered to partners as a package alongside financing and technology.

The strategic danger for the Global North is not that these systems will offer "lower quality" in a technical sense but that they will legitimise emissions without equivalent integrity checks and thereby weaken the bargaining position of the integrity-focused buyers. The danger for the Global South is that it will be divided: those who accept Northern standards may gain access to higher-priced demand, while those who remain outside may find that their credits are consigned to a lower-priced, lower-liquidity segment. The danger to the climate is a "race to the bottom" in which buyers seek the cheapest credits that satisfy regulatory or reputational requirements.

3.4 Assessment: Fragmentation as Equilibrium

A pragmatic assessment suggests that full convergence is neither likely nor necessarily desirable in the near term. Fragmentation has costs: it increases transaction costs, creates arbitrage opportunities, and exposes systems to integrity risk. But it also offers benefits: it permits experimentation, avoids the need to agree on a single price, and allows governments to build domestic legitimacy for carbon pricing before linking. The policy question is therefore how to manage fragmentation: how to establish mutual-recognition mechanisms, minimum integrity floors and interoperable registries without requiring a single global system. The proposals discussed in Chapter 6, including a "Paris Club for carbon" in the form of a recognition framework linked to Article 6, and the principle of "conditional interoperability", respond to this need. The alternative is a competitive dynamic in which standards become instruments of exclusion, and the climate regime is overtaken by trade and security logic.

Chapter 4: Case Study: The EU's CBAM and the Global South's "Climate Justice" Backlash (2023-Present)

4.1 Context: From Reporting Phase to Financial Liability

4.1.1 Timeline of the Transition

The case study opens on 1 October 2023, when the transitional phase began and importers in the six covered sectors became responsible for quarterly reports of embedded emissions. For the Global South, this period was a rehearsal for compliance and a window for diplomacy. The principal milestones were as follows. In August 2023 the BASIC group (Brazil, South Africa, India, China) issued a ministerial statement rejecting unilateral trade measures framed as climate action. Through 2024 the Commission published implementing acts on methodology, default values and verification, and simplified the reporting rules after complaints from importers about burdens. In early 2025 the "Omnibus" proposals introduced the 50-tonne threshold and the later certificate schedule, an acknowledgement that the original administrative design was heavier than the EU's own importers could bear. In November 2025, COP30 in Belém hosted a confrontation over trade measures. On 1 January 2026 the definitive phase began, with the first obligations accruing on 2026 imports. On 1 February 2027 certificate sales open, and on 30 September 2027 the first annual declaration falls due. Free allocation is phased out between 2026 and 2034.

It is important to understand the nature of the 2026 "financial penalty rollout" accurately. No certificate payments are due during 2026; the liability accrues on imports during the year and is settled in 2027. The CBAM factor in 2026 is 2.5 percent, so the effective charge in the first year is modest. The immediate strategic effect of 2026 is therefore not primarily financial; it is the *legal crystallisation* of the regime and the *behavioural effect* on contracts, investment and diplomacy. Exporters now have to fix their data and verification arrangements; importers in the EU have to obtain authorised-declarant status; and governments in exporting countries have to decide whether to pursue recognition, retaliation or accommodation.

4.1.2 The Exposure Map

The exposure of exporting countries varies significantly with product mix, destination share and carbon intensity. India's exposure arises from steel, aluminium and fertilisers: the EU is one of its most important destinations for iron, steel and aluminium products, and its coal-heavy grid raises embedded emissions. South Africa's exposure arises from ferrochrome, steel, aluminium and fertilisers, where electricity intensity is high because the grid remains dominated by coal. Brazil's exposure is lower for steel and aluminium, whose grid is predominantly renewable, but significant for specific exporters in iron ore pellets, semi-finished steel and fertilisers; the Brazilian position is therefore driven by principle and by its ambition to lead the Southern coalition rather than by acute trade exposure. China's exposure is large in absolute terms but a modest share of its total exports, and it has substantial capacity to respond through domestic pricing and by diverting products to other markets. The most acute vulnerabilities are among smaller states with a narrow export base: Mozambique (aluminium), Zimbabwe (ferro-alloys), Ukraine, Georgia, Egypt and Turkey. These countries have the least capacity for MRV and the greatest need for technical assistance.

4.2 The Southern Critique: Climate Justice Reframed as Trade Doctrine

4.2.1 The Intellectual Foundation

The Southern critique is built on three doctrinal pillars. The first is **CBDR-RC**, enshrined in Article 3.1 of the UNFCCC and Article 2.2 of the Paris Agreement. It holds that parties should act "on the basis of equity and in accordance with their common but differentiated responsibilities and respective capabilities". Developing states argue that CBAM, by applying a uniform carbon-cost methodology to all exporters, negates differentiation. The EU responds that CBAM does not differentiate between producers on the basis of nationality but on the basis of the carbon price they pay, and that CBDR-RC governs the obligations of states, not the regulation of imports by importing states.

The second pillar is **UNFCCC Article 3.5**, which provides that measures taken to combat climate change, "including unilateral ones", should not constitute "a means of arbitrary or unjustifiable discrimination or a disguised restriction on international trade". Southern states invoke this provision as the specific textual basis for challenging CBAM in the climate regime; the EU's counter-argument is that CBAM is non-discriminatory in design and thus within the permitted category. The third pillar is **historical responsibility**: cumulative emissions since 1850 are dominated by the United States, EU and other industrialised countries, and developing states argue that they are being asked to accept the costs of a problem that others created. The "climate justice" frame, which resonates well beyond government, links this argument to moral language and to civil-society networks that give the Southern position a legitimacy that a purely commercial objection would lack.

4.2.2 Declaratory Positions

India has been the most vocal and the most legalistic critic. Its government has called CBAM a unilateral measure that violates CBDR-RC and WTO principles, has argued in the Council for Trade in Goods and in bilateral talks that it should be exempted or compensated, and has raised the prospect of reciprocal measures. South Africa has made the case primarily as a development argument: carbon-intensive electricity is the consequence of an energy-system history it is seeking to change with international support through its JETP, and CBAM penalises a transition that is under way but incomplete. Brazil has taken the role of institutional entrepreneur. Brazil's COP30 presidency proposed mechanisms to review unilateral trade measures within the UN climate process, a proposal the EU resisted because it would subject CBAM to a venue where it commands no majority. China has issued strongly worded rejections in the WTO and at COP meetings, while quietly building the national ETS that would enable it to claim deductions. In a revealing divergence, the African Group and the LDC Group have combined opposition with a pragmatic request for exemption or technical and financial support, a position reflecting both their vulnerability and their limited retaliatory capacity.

4.2.3 The COP30 Confrontation

At COP30, developing countries including China, India and Saudi Arabia criticised CBAM as protectionist and as introducing unfair trade barriers, and Brazil proposed regular UN reviews of such measures. The EU strongly opposed, arguing that trade disputes should be resolved in trade fora and not climate negotiations. The outcome was a procedural compromise, with an agreement to hold dialogues on the trade dimensions of climate action in the subsidiary-body process rather than a binding mechanism. This outcome is analytically important because it demonstrates the "forum-shopping" dynamics at the heart of the dispute: the South seeks to relocate the controversy to the UNFCCC, where it has a numerical majority and can invoke equity principles; the EU prefers the WTO, where legal arguments are technical, procedures are slow and the Appellate Body is non-functional.

4.3 Retaliation Strategies: The WTO Toolkit

4.3.1 The Structural Constraint: A Paralysed Appellate Body

The WTO dispute-settlement system has been unable to complete appeals since December 2019, when the US blocked appointments to the Appellate Body. A panel ruling against a respondent can be appealed into the void, so a formal "win" may not be enforceable. This changes the strategic calculus of both complainant and respondent. For complainants, a panel report remains valuable as a legitimacy instrument even if it cannot be enforced; for respondents, the absence of a binding appeal reduces the cost of an adverse ruling. The interim arrangement among a subset of members (the Multi-Party Interim Appeal Arbitration Arrangement, MPIA), in which the EU participates and Brazil and China also take part, offers a limited alternative, but India and South Africa are not parties, which complicates any dispute involving them with the EU.

4.3.2 India's Options

India's available instruments can be divided into the formal (WTO), the quasi-formal (notifications under the Agreement on Safeguards) and the bilateral (trade negotiations). In the first category, India could initiate a consultations request under GATT Article XXII and the Dispute Settlement Understanding, alleging inconsistency with Articles I and III of the GATT and the Agreement on Technical Barriers to Trade, and contesting the availability of Article XX. In the second, India has pioneered an unconventional approach: in 2025 it notified the WTO that it reserved the right to retaliate against US steel and aluminium tariffs by invoking Article 8 of the Agreement on Safeguards, which permits the suspension of concessions in response to safeguard measures; it has indicated willingness to deploy similar reasoning against other measures it regards as protectionist, and has stated that it reserves the right to respond to the UK CBAM. The legal strength of this approach is open to challenge, since the EU and UK maintain that CBAM is not a safeguard measure, but the signalling value is high. In the third category, India has used bilateral negotiations as its principal leverage: during the EU-India free trade negotiations that concluded in January 2026, CBAM was among the most contentious points, alongside steel and automobiles. India secured a dedicated climate chapter and a statement of understanding on recognition of its future carbon market, but no exemption, and the Commission publicly emphasised that India would receive no more favourable treatment than other partners.

India's preference ranking appears to be: negotiate recognition of its CCTS under Article 9; build the CCTS so that its price signal is credible; accept CBAM as a fact while seeking compensation through technology and finance; and preserve retaliation as a latent threat. The decision not to launch a WTO case immediately reflects a calculation that litigation would harden positions and jeopardise the trade agreement, while its signalling statements maintain pressure.

4.3.3 Brazil's Options

Brazil's strategy is institutionally distinct. Its exposure is limited, its relations with the EU are complicated by the EU-Mercosur agreement (concluded in principle in late 2024 and subject to ratification politics), and it has more to gain from the role of broker. Its approach has three elements: (1) **multilateral reframing**, using the COP30 presidency to bring trade measures into the UNFCCC agenda; (2) **legal positioning**, preserving the right to bring a WTO challenge, supported by its record as a frequent and

successful litigant in earlier trade disputes (notably against US cotton subsidies and EU sugar policies); and (3) **reciprocal domestic measures**, including the SBCE and the Brazilian "eco-invest" programme, which seek to ensure that carbon revenue is retained domestically. Brazil has also drawn on its leverage in the EU-Mercosur negotiations: European agricultural and environmental interests must be weighed against Brazilian objections to carbon-related provisions, particularly deforestation rules and CBAM.

4.3.4 South Africa, China and the Coalitions

South Africa faces the most direct trade-off between principle and pragmatism. It is a member of the WTO and the BASIC group, and its government has supported collective critiques of CBAM, yet its economy is deeply integrated with Europe and its negotiators have put priority on securing finance via the JETP and on accessing technical assistance. China, as the largest exporter in covered categories, has the capacity to bring a case and the retaliatory leverage of its mineral-export controls, but it has so far preferred to use rhetoric and bilateral channels, partly because it has little interest in strengthening WTO norms that could also constrain its own subsidies. The coalition dynamic is therefore one of **shared rhetoric but divergent strategies**: a joint statement is easy; a joint WTO case is hard, because the members' interests differ and because each fears that a legal victory could set precedents that bind its own policies.

4.3.5 The Russian Case and Its Signal

Russia's request for consultations in 2025 initiated the first formal WTO proceeding on CBAM. Russia is a marginal player in CBAM-covered exports to the EU after the post-2022 sanctions, so the case is largely symbolic, and the political reality that the first complainant is a sanctioned state complicates the optics for other Southern states, which have little desire to appear aligned with Moscow. Nonetheless, the proceeding creates a legal vehicle: third parties (India, Brazil, China, South Africa) can join, and a panel report would create a body of reasoning on the central legal questions. The EU has a defensive interest in ensuring that any adjudication is conducted in a forum where it is better placed to prevail.

4.4 The Western Strategic Pivot: Climate Clubs, Just Transition Partnerships and Conditional Modernization

The EU and its partners recognised that a purely legal or technical defence of CBAM would not be sufficient to prevent a diplomatic rupture, particularly when the US is no longer a partner on climate and China is positioned as an alternative. The response has been to supplement the stick of CBAM with carrots, giving rise to a layered strategy that combines three instruments.

4.4.1 Climate Clubs

The G7 Climate Club, launched in December 2022 under the German presidency and now comprising around three dozen members including the EU, aims to accelerate decarbonisation of heavy industry through common methodologies for measuring the greenhouse-gas intensity of products, transparent pathways, and cooperation on technology and finance. The Club's intent is to align the measurement frameworks that CBAM depends on, so that products made in member states, including emerging economies that have joined, are measured in compatible ways. Nordhaus's original conception of a club envisaged tariffs on outsiders; the Western version is an open club with an emphasis on inclusion and capacity building. The tension is that the club presents itself as cooperative while operating alongside a border measure whose function is to penalise those who stay outside; for Southern governments, the invitation to join looks like an invitation to accept the standard-setters' rules.

The strategic design problem is membership. A club that includes major emerging economies has greater legitimacy but diluted standards; a club of like-minded states is cohesive but reinforces the perception of a Northern bloc. The loss of the United States as an active participant after 2025 reduces its weight, leaving the EU, UK, Japan, Canada and a number of smaller members to carry the initiative. Alternative formats, such as sectoral clubs for steel, cement and aluminium, and a prospective "buyers' club" for green steel demand, have been proposed, with the aim of creating lead markets for low-carbon products and thereby reducing the cost of compliance.

4.4.2 Concessional Just Transition Partnerships

The JETPs, discussed in Chapter 1, are the second instrument. Conceived as bilateral packages of grants, concessional loans, guarantees and policy commitments, they are explicitly intended to enable coal retirement and renewable deployment in countries with large fossil-based power sectors. In the CBAM context, their strategic function is to demonstrate that Northern states will help countries meet the standard CBAM imposes. The EU's complementary instruments include the Global Gateway investment programme, the Clean Industrial Deal (February 2025), and the proposed Clean Trade and Investment Partnerships (CTIPs), which would offer partner countries preferential access, investment support and technical assistance in exchange for cooperation on clean-technology value chains and supply security. The EU has also committed, in the CBAM regulation itself, to technical assistance for developing countries, and has proposed that a portion of CBAM revenue be directed to the EU budget as an "own resource"; the allocation of any share to a facility for LDCs, long demanded by the African Group, remains politically unresolved.

The effectiveness of the JETP model is, as discussed, constrained by delivery shortfalls and the withdrawal of the US. A fundamental tension also arises from the conditionality structure: partner governments are asked to accept coal-retirement and policy-reform milestones as the price of access to finance, but the finance is predominantly in loans and the timeline for disbursement is slower than the timeline of CBAM obligations. The risk is that exporting countries will bear the CBAM charge before the Northern money arrives.

4.4.3 Conditional Modernization Funding

The third instrument is the offer of funding conditional on industrial modernisation, including support for green-steel, green-hydrogen and fertiliser projects that are intended to reduce embedded emissions and secure future access to the EU market. Bilateral examples include EU investments in hydrogen and critical-materials partnerships with countries in Africa, Latin America, Central Asia and the Gulf, and European development finance institutions' lending to decarbonise industrial plants in partner states. These offers carry a strategic logic: they convert the costs of CBAM compliance into investment flows that benefit European technology suppliers and secure supply chains for European industry, so that decarbonisation in exporting countries is financed in part by Europe and serves Europe's strategic interests. The question for recipient governments is whether this represents partnership or tied aid. For a state such as India, the offer is attractive if it provides low-cost capital for steel decarbonisation but unattractive if it locks the supply chain into European equipment and conditions. The distinctive feature of the "conditional" model is that the same state may be both beneficiary and target of the policy.

4.4.4 The Combined Effect

The combined effect of these instruments is to transform a potentially unilateral confrontation into a differentiated bargaining process. The EU has not offered a general exemption, which would defeat CBAM's purpose and violate the WTO principle of non-discrimination. It has instead offered individualised accommodation, whether through free-trade-agreement chapters, recognition of carbon pricing, finance, or technology cooperation. This produces a hub-and-spoke diplomacy in which each Southern state negotiates separately with the EU, weakening the cohesion of the Southern coalition. The tactic has been effective in the short run: despite the rhetoric, no state has so far imposed retaliatory measures specifically in response to CBAM, and several (India, Indonesia, Turkey, Brazil) are building the domestic carbon pricing that CBAM is designed to induce. Whether the approach is sustainable depends on whether the promised support materialises on a timeline commensurate with the burden.

4.5 Analysis and Scenarios to 2030

4.5.1 Assessment of the Strategic Outcome

Three assessments emerge from the case study. First, **CBAM has been effective as a lever** of regulatory diffusion, whatever one concludes about its fairness: carbon-pricing initiatives have accelerated in precisely the jurisdictions most exposed to it. Second, **its legitimacy deficit is real and consequential**, because the EU's sustained legitimacy as a climate leader depends on Southern consent, which it has not secured. The EU's attempt to confine the debate to trade fora is rational legally but costly diplomatically. Third, **the decisive variable is the Global North's credibility on finance**. A mechanism that penalises developing-country producers for carbon intensity will be accepted as fair only if the same countries have credible access to the capital and technology needed to decarbonise. The persistence of the finance gap is therefore the principal threat to CBAM's long-term stability.

4.5.2 Scenarios

Table 4.1: Scenarios for the CBAM-Global South relationship to 2030 (analytic judgement, not forecast)

Scenario	Description	Leading indicators	Relative likelihood
A: Managed accommodation	CBAM stays intact; rules-based recognition of foreign carbon prices; a support facility for LDCs and lower-middle-income exporters; WTO dispute remains symbolic	Recognition decisions for India, Brazil and others; revenue earmarked for partners	Highest
B: Grudging compliance	Exporters comply and build domestic ETSs; rhetoric persists; finance gap unresolved; litigation limited	Spread of domestic carbon prices; continued COP trade-dialogue stalemate	High (often blended with A)
C: Escalation	A panel case or tit-for-tat measures; spillover into critical-mineral export controls or other trade domains	Formal retaliation notices; mineral licensing tightened	Low to moderate (tail risk)
D: Dilution	EU narrows scope or extends delays under industrial lobbying and trade-deal pressure; credibility of the price signal weakens	Further postponements; wider exemptions in trade agreements	Moderate

Our central expectation is Scenario A or a blend of A and B. The structural factors are as follows: the EU is committed to the regime but responsive to administrative simplification; exporting governments have a strong interest in capturing carbon revenue domestically; and no Southern government wishes to sacrifice access to the European market. The tail risk is Scenario C, in which a

dispute over a specific measure (for instance, an extension to downstream products, a strict approach to default values, or the conspicuous exclusion of a major partner's carbon market from recognition) triggers retaliation that spills into other domains, such as critical-mineral export controls.

Chapter 5: The Paris Agreement Structured Matrix

5.1 Analytical Framework

The Paris Agreement of 2015 established a universal architecture in which every party submits nationally determined contributions (NDCs) on a five-year cycle, with a collective temperature goal of well below 2 degrees Celsius and efforts toward 1.5 degrees, a transparency framework, a global stocktake, and the finance and market provisions examined in earlier chapters. Its legal character is distinctive: binding procedural obligations coexist with non-binding substantive targets. This design maximised participation at the cost of enforcement, and means that each party's evaluation of the Agreement reflects a calculation about what it provides in security, competitiveness and diplomatic position rather than a compliance obligation.

This chapter assesses the Agreement from the standpoint of five actors: the United States, the European Union, China, India, and the Least Developed Countries (LDCs). For each, Matrix 5.1 identifies the geopolitical pros, the economic cons and the strategic position as of October 2026, and Matrix 5.2 summarises the net balance on the three axes specified in the project framework: sovereign energy security, economic competitiveness and diplomatic alignment. The assessments are analytic judgements rather than measured quantities; they are intended to make the structure of each actor's interests explicit.

5.2 Matrix 5.1: Geopolitical Pros, Economic Cons and Strategic Positions

Matrix 5.1: Geopolitical pros, economic cons and strategic positions (as of October 2026)

Actor	Geopolitical pros	Economic cons	Strategic position
United States	Historic architect of the bottom-up design and a leading voice in standards; convening power through finance and technology; flexibility of nationally determined targets suited US preferences	Perceived external constraint on energy policy; domestic costs of decarbonisation without reciprocal action by competitors; politically contested finance obligations	Formally withdrawn since 27 January 2026 (second exit); pursues energy-export dominance (LNG), tariffs and bilateral deals; risks ceding clean-tech markets and rule-setting to the EU and China
European Union	Platform for regulatory leadership; validates the EU model; exports ETS and CBAM standards; reduced fossil-import dependence after 2022	Highest domestic carbon and energy costs among major economies; leakage and competitiveness pressure against China and the US; large clean-tech import dependence	Seeks to keep Paris alive: 2035 NDC range of roughly 66-72.5% and 90% net cut by 2040; Climate Clubs, CBAM and partnership diplomacy; leading finance provider but under fiscal strain
China	Claims responsible-power legitimacy without binding obligations; leadership vacuum after US exit; global market for its clean technology; electrification cuts oil-import exposure	Coal dependence and rising absolute emissions until peak; overcapacity backlash and trade remedies; large CBAM exposure in absolute terms	Peak before 2030, neutrality before 2060; 2035 NDC of 7-10% below peak; retains developing-country status; South-South positioning; national ETS expanding toward CBAM recognition
India	Codifies differentiation (CBDR-RC) and finance claims; Global South leadership; clean-energy manufacturing and lower fuel-import bill; International Solar Alliance leadership	Coal-heavy power system; CBAM exposure in steel, aluminium and fertilisers; high cost of capital; limited fiscal space	Net zero 2070; 500 GW non-fossil capacity by 2030; CCTS in implementation; resists binding sectoral phase-outs; hedges between the EU (FTA), BRICS and other partners
Least Developed Countries (LDCs)	Equal voice in the only forum where they hold formal parity; access to adaptation and loss-and-damage finance; moral authority on vulnerability	Finance mostly loans, not grants; weak absorptive capacity; acute CBAM exposure for a few exporters; small share of global emissions (around 1%)	LDC Group of 44 states bargains as a bloc; priorities are grant-based adaptation finance, loss and damage and technical assistance; highly exposed to donor fatigue

5.3 Matrix 5.2: Pros and Cons on the Three Strategic Axes

Matrix 5.2: Net balance by strategic axis (analytic assessment)

Actor	Sovereign energy security	Economic competitiveness	Diplomatic alignment
United States	**Pro:** shale and LNG surplus supports autonomy. **Con:** exit forgoes grid and electrification gains.	**Pro:** lower regulatory cost in fossil sectors. **Con:** risks losing clean-tech export markets.	**Pro:** bargaining freedom. **Con:** diplomatic isolation; loss of credibility on finance.
European Union	**Pro:** renewables and efficiency reduce import exposure. **Con:** new dependence on Chinese equipment and minerals.	**Pro:** first-mover clean industry; ETS revenue. **Con:** high energy and carbon cost.	**Pro:** norm-setting authority. **Con:** legitimacy deficit in the South over CBAM and finance.
China	**Pro:** electrification and domestic supply chains cut oil-import exposure. **Con:** coal reliance persists.	**Pro:** manufacturing scale and cost leadership. **Con:** overcapacity and trade remedies.	**Pro:** leader of Southern coalition. **Con:** pressure to accept contributor obligations.
India	**Pro:** solar and domestic manufacturing reduce fuel-import bill. **Con:** coal remains the reliability backbone.	**Pro:** cost-competitive renewables. **Con:** CBAM and capital-cost penalties on heavy industry.	**Pro:** multi-aligned broker. **Con:** exposed to being squeezed between blocs.
LDCs	**Pro:** decentralised renewables can leapfrog grids. **Con:** reliance on external capital and equipment.	**Pro:** limited trade exposure overall. **Con:** adaptation costs and debt burden.	**Pro:** coalition voice and moral standing. **Con:** minimal retaliatory or bargaining capacity.

5.4 Interpretation of the Matrix

5.4.1 Divergence of Interests Beneath a Common Treaty

Three patterns emerge. The first is that **no actor's interest in the Agreement is purely environmental**. The EU values the Agreement as a platform for regulatory leadership and as validation of a decarbonisation model it has already paid for. China values it as a venue for claiming systemic responsibility without binding constraints. India values the principle of differentiation and the finance commitments. The LDCs value it as the only forum in which they hold equal voice. The United States, under the current administration, regards it as a constraint on a fossil-energy strategy and a vehicle for foreign influence over domestic policy.

The second is **asymmetric exposure to enforcement by other means**. The Agreement has no mechanism to penalise non-compliance, but its principles are being enforced by trade measures (CBAM) and finance conditionality, which fall disproportionately on mid-income exporters and on recipients of bilateral finance. The Agreement thus has a *shadow enforcement regime* controlled by those with market and financial power, and this regime has a more visible effect on the Southern states than the Agreement's own transparency framework. The third is the **impact of the US exit**. The formal withdrawal on 27 January 2026, and the signalled intention to leave the UNFCCC itself, removes the largest historical emitter and one of the largest finance providers. In the near term this strengthens the relative position of the EU and China as rule-setters, and reduces the credibility of the finance pledges that sustained the bargain with developing countries. In the longer term it creates the possibility of a "US-less" climate order in which border measures like CBAM become a primary means of pressure on those who remain outside, and indeed a tool that could be directed at the US itself.

5.4.2 The Position of the Middle Powers

The matrix does not capture the large group of middle powers: Brazil, Indonesia, South Africa, Turkey, Saudi Arabia, Mexico, Vietnam and others. These are the "swing" states whose alignment will be shaped by whichever package of finance, market access and technology is most credible. Their positions typically combine the India-type argument about differentiation with a pragmatic acknowledgement of the economic gains from clean-energy manufacturing and carbon-market participation. Their aggregate behaviour will determine whether the Paris architecture remains a single system or becomes one of several competing orders.

Chapter 6: Strategic Outlook and Policy Recommendations

6.1 Strategic Outlook: Five Structural Trends to 2035

Trend 1: Climate-trade convergence. The boundary between climate policy and trade policy has dissolved. Border carbon adjustments, clean-technology subsidies, critical-mineral export controls and carbon-credit eligibility rules are already instruments of foreign economic policy. By 2035 most large economies will operate some form of border adjustment or clean-product procurement preference, and the main question will be whether these are coordinated or conflicting.

Trend 2: Finance as the binding constraint. The scale of the transition required in emerging and developing economies, in the order of trillions of dollars annually, cannot be met by public concessional finance alone, so the main geopolitical contest will be over the conditions under which private and institutional capital reaches those economies: the cost of capital, currency risk, and the credibility of guarantees. Reform of the multilateral development banks, debt restructuring frameworks and country platforms will matter more than headline pledges.

Trend 3: Technology-sovereignty competition. The concentration of clean-technology supply chains in China will continue to drive diversification efforts in the West and in India, Indonesia and others, but diversification will be slow, expensive and partial. Export controls on critical inputs will remain bargaining instruments, as will Western controls on advanced technologies. Mutual vulnerability will persist.

Trend 4: Fragmented carbon pricing with selective interoperability. Compliance carbon pricing will spread (China, India, Brazil, Indonesia, Turkey, the UK and others) with uneven stringency, and bilateral recognition agreements will multiply. A single global price is not feasible within the decade; a network of recognised systems is.

Trend 5: Legitimacy as strategic capital. Measures perceived as unjust will generate counter-coalitions, as the history of CBAM demonstrates. States and blocs that combine stringent standards with visible, reliable support will accumulate legitimacy; those that do not will face resistance at every venue.

6.2 Policy Recommendations

6.2.1 For the European Union and Likeminded Governments

R1: Sequence CBAM obligations with finance delivery. The EU should establish a transparent link between the CBAM factor schedule and the delivery of support to exporting countries. A “CBAM Transition Facility” capitalised from a defined proportion of CBAM revenue, with priority for LDCs and lower-middle-income exporters, would address the most acute equity objection at the lowest political cost. Its design should emphasise grants and concessional instruments for MRV capacity, and for decarbonisation investments in covered sectors. The risk of a fiscal-neutrality objection (CBAM revenue is increasingly earmarked for the EU budget) should be addressed by ring-fencing a modest fraction, with reporting.

R2: Institutionalise mutual recognition. The Commission should publish clear, rules-based criteria for recognising foreign carbon prices and MRV systems under Article 9, a timetable for decisions, and an appeal or review procedure. Predictability of recognition is the best incentive to adopt domestic carbon pricing and the best defence against the charge of arbitrariness. Recognition decisions on India, Brazil, Indonesia and Turkey should be sequenced to reward credible system design rather than diplomatic alignment.

R3: Calibrate default values and phase them in with verification support. Default values should be set at levels that incentivise measurement without punishing producers who lack verifiers. The EU should finance accreditation capacity in exporting countries, and accept verification by accredited local bodies under common standards.

R4: Defend CBAM in the WTO while negotiating an interpretive understanding. Rather than waiting for a panel ruling that may be appealed into the void, the EU should pursue a ministerial declaration or understanding on the treatment of non-discriminatory border carbon adjustments under Article XX, ideally in cooperation with partners in a plurilateral format. This would reduce legal uncertainty for all parties, including the EU.

R5: Treat the extension of CBAM scope with caution. The proposed extension to downstream products and anti-circumvention provisions increases administrative burdens and the number of affected states. Extension should follow, not precede, evidence that the first phase is administratively manageable and that support commitments are being met.

6.2.2 For Developing and Emerging Economies

R6: Build domestic carbon pricing designed for recognition. Governments that expect to remain exporters of covered goods should design their systems for Article 9 compatibility: clear coverage, verifiable data, an effective price and revenue retention. This converts a perceived imposition into a source of domestic fiscal revenue.

R7: Coordinate on technical positions rather than only rhetoric. The Southern coalition's influence would be strengthened by a joint technical proposal on default values, MRV capacity and revenue use, rather than by declarations on principle alone. A shared proposal, including a minimum-viable "common MRV protocol", would be harder for the EU to dismiss.

R8: Preserve market diversification and avoid exclusive dependence. Exporting states should diversify both markets and suppliers for clean technology, using Chinese, European, Japanese, Korean and domestic sources to maintain bargaining leverage.

R9: Capture value in critical-mineral chains. Resource-rich states should develop processing and manufacturing capacity through partnerships with multiple investors, while avoiding export bans that provoke retaliation and that in practice reduce investment. Regional frameworks, such as the African Union's Green Minerals Strategy, offer a vehicle for common standards on environmental, social and governance conditions.

6.2.3 For the International System

R10: A "Paris Club for carbon". Establish a recognition framework, under or alongside Article 6, in which participating carbon-pricing systems commit to minimum integrity floors (baseline, additionality, verification, registry transparency and corresponding adjustments), and in return obtain mutual recognition for purposes such as border adjustment deductions and compliance offsetting. The framework should be open to all systems that satisfy the floors, avoiding the creation of a bloc.

R11: Reform climate finance accounting. The OECD, the UNFCCC and the multilateral development banks should adopt harmonised definitions of "climate-specific" finance, report grant-equivalent values, and publish country-level disbursement data. The NCQG's credibility depends on transparent measurement; otherwise the experience of the USD 100 billion goal will be repeated.

R12: Strengthen the Bridgetown and MDB reform agendas. Lower the cost of capital through expanded MDB lending capacity, hybrid-capital instruments, guarantee platforms, climate-resilient debt clauses and debt-for-climate swaps; link these to country platforms in which recipient governments lead.

R13: Create a trade-climate dialogue with legal standing. Establish a standing joint forum between the UNFCCC and the WTO secretariats (building on the dialogue initiated at COP30) in which climate-related trade measures can be reviewed, with a mandate to recommend adjustments rather than adjudicate disputes.

R14: De-escalate critical-mineral export controls through transparency and diversification. Importing and exporting states should agree on advance-notification norms for export restrictions in critical minerals, in parallel with Western commitments to buy from diversified sources at stable prices through long-term offtake and price-floor mechanisms.

6.3 Implementation Priorities and Risk Register

The recommendations vary in feasibility. Those that lie within a single jurisdiction's authority (R1 to R3, R5, R6) can be implemented within the 2026 to 2028 window, and should be prioritised because the CBAM factor begins to rise steeply after 2029. Those requiring plurilateral agreement (R4, R10, R13, R14) are more complex and should be sequenced through the G20 under the presidency rotation, the WTO's ministerial process and the COP cycle. The principal risks are: (i) donor fatigue and budget competition from defence spending; (ii) an adverse WTO ruling that triggers a political backlash; (iii) a breakdown in US-China relations that triggers wider export-control escalation; (iv) integrity scandals that undermine confidence in carbon markets; and (v) domestic political reversals in major economies that unwind carbon pricing.

6.4 Conclusion

The geopolitics of climate finance, carbon markets and the green transition is neither a morality play nor a purely technical negotiation over emissions accounting. It is a structural contest over who finances the transition, who sets the standards by which it is measured, and who captures the industrial rents it creates. The evidence reviewed here supports three conclusions. The Global North's leverage has been weakened by unreliable finance delivery but retains the power of standard-setting and market access, as CBAM demonstrates. The Global South's leverage lies less in collective legal action than in control of critical resources, large domestic markets and the ability to build alternative institutions, with China as its most capable but least disinterested partner. And the most likely outcome is neither convergence nor rupture, but an asymmetric, negotiated fragmentation whose stability will depend on whether the parties can build mutual-recognition mechanisms before distrust hardens into economic blocs. Climate policy is the new arena for great-power competition; the central task for statecraft is to ensure it remains an arena in which competition and cooperation can coexist.

Note on Sources, Data and Limitations

This report draws on publicly available primary and secondary sources, including OECD reporting on climate finance, UNFCCC decisions (Copenhagen 2009, Cancún 2010, Paris 2015, Baku 2024, Belém 2025), the text of Regulation (EU) 2023/956 and its 2025 amendments, European Commission CBAM guidance, IEA analyses of critical minerals and clean-energy investment, UNCTAD and UNEP assessments, WTO legal texts and jurisprudence, and contemporaneous reporting on COP30, the EU-India trade negotiations and US withdrawal from the Paris Agreement. Quantitative values are given as approximations where methodologies vary; the OECD financial series in Figure 1.1 reflects figures as originally published and is subject to revision. Several developments (recognition decisions, WTO proceedings, trade-agreement texts) are evolving, and the reader should verify current status against primary sources before relying on them in formal submissions. The Matrix assessments in Chapter 5 and the scenarios in Chapter 4 are analytical judgements, not forecasts. This report does not constitute legal advice.